Exercise Sheet 18

The problems use the notation of Lecture 18: $T^d = (\mathbb{C}^n)^{\otimes d}$, $\bar{\lambda} = \sum_i (\lambda_i - \lambda_{i+1})\omega_i$, and $\det = \Lambda^n \mathbb{C}^n$. Exercises 18.1–18.3 practise labels, exterior powers and dimensions. Exercises 18.4–18.6 reconstruct and transfer the lecture's arguments.

Exercise 18.1 (Partitions and Dynkin labels for $\mathfrak{sl}_4$).

(i) Find the Dynkin labels and the normalised partition of $\mathbb{S}_\lambda(\mathbb{C}^4)$ for $\lambda = (3,1,1)$, $(2,2)$, $(4,2,1,1)$ and $(2,2,2,2)$.

(ii) Find normalised partitions for $L(\omega_1 + 2\omega_3)$, $L(\omega_2)$ and the adjoint module $L(\omega_1 + \omega_3)$.

(iii) Let w_0 be the longest element of S_n. Prove $-w_0\omega_i = \omega_{n-i}$. Let λ be a partition with $\ell(\lambda) \le n$ and put $\lambda^c_j = \lambda_1 - \lambda_{n+1-j}$. Deduce from the dual formula $L(\nu)^* \cong L(-w_0\nu)$ of Lecture 14 that $\mathbb{S}_\lambda(\mathbb{C}^n)^* \cong \mathbb{S}_{\lambda^c}(\mathbb{C}^n)$ as $\mathfrak{sl}_n$-modules. Show that over $\mathfrak{gl}_n$, $\mathbb{S}_\lambda(\mathbb{C}^n)^* \cong \mathbb{S}_{\lambda^c}(\mathbb{C}^n) \otimes \det^{-\lambda_1}$. Compute the dual of $\mathbb{S}_{(3,1,1)}(\mathbb{C}^4)$ over $\mathfrak{sl}_4$ and over $\mathfrak{gl}_4$.

Exercise 18.2 (Exterior fundamental modules). Let $1 \le k \le n$ and write $e_J = e_{j_1} \wedge \cdots \wedge e_{j_k}$ for $J = \{j_1 < \cdots < j_k\}$. Write $\rho_{\Lambda^k}(X)$ for the induced action on $\Lambda^k\mathbb{C}^n$:

$$\rho_{\Lambda^k}(X)(v_1 \wedge \cdots \wedge v_k) = \sum_{j=1}^{k} v_1 \wedge \cdots \wedge Xv_j \wedge \cdots \wedge v_k.$$

(i) Prove $\mathbb{S}_{(1^k)}(\mathbb{C}^n) \cong \Lambda^k\mathbb{C}^n$ as $\mathfrak{gl}_n$-modules.

(ii) For $k \le n-1$, prove directly that $\Lambda^k\mathbb{C}^n$ is an irreducible $\mathfrak{sl}_n$-module with highest vector $e_1 \wedge \cdots \wedge e_k$ of weight ω_k.

(iii) Recover $\dim \Lambda^k\mathbb{C}^n = \binom{n}{k}$ from the Weyl dimension formula.

(iv) Prove $\Lambda^{n-k}\mathbb{C}^n \cong (\Lambda^k\mathbb{C}^n)^* \otimes \det$ as $\mathfrak{gl}_n$-modules, and hence $\Lambda^{n-k}\mathbb{C}^n \cong (\Lambda^k\mathbb{C}^n)^*$ over $\mathfrak{sl}_n$.

Exercise 18.3 (The hook-content formula). For a box $b = (i,k)$ of λ, its content is $c(b) = k - i$ and its hook length is $h(b) = (\lambda_i - k) + (\lambda'_k - i) + 1$, where λ' is the conjugate partition. Let $\ell(\lambda) \le n$ and $l_i = \lambda_i + n - i$.

(i) Deduce from the Weyl dimension formula of Lecture 17 that

$$\dim \mathbb{S}_\lambda(\mathbb{C}^n) = \prod_{1 \le i < j \le n} \frac{l_i - l_j}{j - i}.$$

(ii) Fix i. Prove that the hook lengths in row i and the numbers $l_i - l_j$ for $j > i$ together form $\{1, \dots, l_i\}$, without repetition.

(iii) Prove

$$\dim \mathbb{S}_\lambda(\mathbb{C}^n) = \prod_{b \in \lambda} \frac{n + c(b)}{h(b)}.$$

(iv) Check the formula for $(2,1)$ with $n = 3$ and for $(2,2)$ with $n = 4$.

Exercise 18.4 (Highest vectors in $(\mathbb{C}^3)^{\otimes 3}$). Let $n = d = 3$ and $e_{ijk} = e_i \otimes e_j \otimes e_k$.

(i) Find all vectors of weight $2\varepsilon_1 + \varepsilon_2$ killed by $\rho_3(E_{12})$ and $\rho_3(E_{23})$.

(ii) Show that this space is S_3-stable and isomorphic to $V^{(2,1)}$.

(iii) Explain the answer to (ii) by Schur–Weyl duality. Use that the vectors of $L(\nu)$ killed by $\mathfrak{n}_+$ form the highest-weight line, and prove this fact.

(iv) For the tableau $\begin{smallmatrix} 1 & 3 \\ 2 & \end{smallmatrix}$, compute $u' = (1+(13))(1-(12))e_{121}$. Show that u' and the vector u of the lecture form a basis of the space in (i).

(v) Compare $\dim V^{(2,1)}$ with the multiplicity of weight 0 in $\mathbb{S}_{(2,1)}(\mathbb{C}^3)$ as an $\mathfrak{sl}_3$-module. Explain why both are 2.

Exercise 18.5 (Determinant twists and the centre). For part (iv), use the representation $X \mapsto \operatorname{tr}(X)N$ on $\mathbb{C}^2$, where

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

(i) Prove $\mathbb{S}_{(k^n)}(\mathbb{C}^n) \cong \det^k$ for integers $k \geq 1$.

(ii) Prove $(\mathbb{C}^n)^* \cong \Lambda^{n-1}\mathbb{C}^n \otimes \det^{-1}$ over $\mathfrak{gl}_n$. Show that $(\mathbb{C}^n)^*$ is not a summand of any T^d, $d \geq 1$, although it is isomorphic to $\mathbb{S}_{(1^{n-1})}(\mathbb{C}^n)$ over $\mathfrak{sl}_n$.

(iii) For $s \in \mathbb{C}$ let $\mathbb{C}_s$ be the one-dimensional $\mathfrak{gl}_n$-module $X \mapsto s \operatorname{tr} X$. Prove that $\mathbb{C}_s \cong \mathbb{S}_\lambda(\mathbb{C}^n) \otimes \det^k$ for some partition λ with $\ell(\lambda) \leq n$ and $k \in \mathbb{Z}$ only if $s \in \mathbb{Z}$.

(iv) Show that the module $X \mapsto \operatorname{tr}(X)N$ above is not a direct sum of modules $\mathbb{S}_\lambda(\mathbb{C}^n) \otimes \det^k$.

Exercise 18.6 (Two variables). Let $n = 2$ and let $a \geq b \geq 0$ be integers.

(i) Compute $s_{(a,b)}(x_1, x_2)$ from the bialternant.

(ii) Deduce $\mathbb{S}_{(a,b)}(\mathbb{C}^2) \cong V_{a-b}$ over $\mathfrak{sl}_2$ and $\mathbb{S}_{(a,b)}(\mathbb{C}^2) \cong \operatorname{Sym}^{a-b}\mathbb{C}^2 \otimes \det^b$ over $\mathfrak{gl}_2$.

(iii) Decompose $(\mathbb{C}^2)^{\otimes 3}$ by Schur–Weyl duality and compare with the Clebsch–Gordan rule of Lecture 6.